

ORUN - A Virtual reality serious-game for kinematics learning

Jhasmani Tito *
University of Campinas, Brazil

Tania Basso†
University of Campinas, Brazil

Regina Moraes‡
University of Campinas,
Brazil
University of Coimbra,
Portugal

ABSTRACT

Virtual Reality (VR) is one of the new educational technologies that has an imperative importance due to the possibilities that it offers, such as bringing hands-on experiences for learning physics phenomena. This demo report is a serious-game, based on VR, and focused on teaching and learning of specific concepts of kinematics. The game is intended to deliver an immersive experience in which the student has an active role and whose game design includes theoretical concepts to maintain engagement throughout the tasks.

Index Terms: Serious Games—Physics—Virtual Reality—STEM;

1 INTRODUCTION

On the last decade STEM (Science, Technology, Engineering and Mathematics) education has gained relevance due to the increasing demand of future professionals who deal with the challenges of new technologies of our present and future world [2]. Nevertheless, the learning of STEM courses at schools has still been a complex task for students who requires better teaching methods and tools to achieve better learning outcomes.

In this scenario, one of the broadly known advantages offered by the teaching of STEM education is related to the experimentation on teaching and learning science to support more complex mental processes, such as understand abstract concepts, solve problems, and representation of physics-mathematical graphs. Sometimes, however, these activities are not conducted in the best way, being limited to perform just non-immersive demonstrations and results verification. According to Etкина et al. [1], although there are two pedagogical techniques for demonstrations (i.e., (a) students as observers and the teacher demonstrating and explaining what happened; (b) the most effective in which students making predictions before the experiment to verify them during the experimentation), their implementation depends on many conditions. Examples of conditions for implementation are teacher training, equipment and available materials. Therefore, virtual tools, simulators, remote labs has been implemented in education to enhance the possibilities of accessing a better experience with less resources. In this context, using VR on STEM education has created important opportunities for students, such as the control of learning materials, hands-on tasks and immediate feedback on their actions during the execution time [3].

To overcome these challenges and opportunities, we have developed a VR application as a serious-game focused on the learning of kinematics concepts as part of STEM education. The game consists of a immersive environment with objects, tasks and animations, aligned to a game narrative with visual elements that support visualization of physics phenomena.

The next section presents the materials and methods used to implement the game, followed by Section 3, which summarizes our

*e-mail: jhasmanictt@gmail.com

†e-mail: taniab@unicamp.br

‡e-mail: regina@ft.unicamp.br, remoraes@dei.uc.pt



Figure 1: Student playing a game on level 2.

conclusions and future works.

2 MATERIALS AND METHODS

This section is divided in two subsections: Virtual Environment Features, with technical information about the game engine, devices and its features and Game Design, showing relationship between game design elements, activities and learning goals.

2.1 Virtual Environment Features

The application was developed under game engine Unreal Engine 4 (UE v4.26)¹, that has plugins and tools ready to integrate with VR devices, and an embedded API to physics simulations. For a better graphic performance, all 3D models were modeling in low-poly. We use Oculus Rift S² as a HMD (Head Mounted Display), with inside out tracking available and 6 DOF (Degree of Freedom), and 88 degrees of FOV (Field of Vision). For interaction, it was used the Oculus Touch Controller with full-motion tracking and 6 DOF. The first release was developed for Windows-PC, a second release is being adapted for Android to run on Oculus Quest 2 in a stand alone mode.

2.2 Game Design

For this game, we followed the Theory of Digital Games, from Ralph and Monu [4], considering game elements such as game mechanics, narrative, technology, among others. Moreover, we interrelated them with the learning goal to achieve a good learning, engagement and immersion. Firstly, the game mechanics were thought to be as most intuitive as possible and the interaction with objects and tasks taking place in a small area. Thus, moves and navigation occurs in natural way, allowing the player to walk naturally, interacting with the objects using his/hers hands (see Fig. 1). Touch controllers have similar functions as hands and are used to take objects or as a pointer, i.e., a virtual laser on virtual boards.

The main artifact in the game is a cannon, which must be used to shoot spheres and hit specific targets. Also, this is the main game

¹<https://www.unrealengine.com/>

²<https://www.oculus.com>

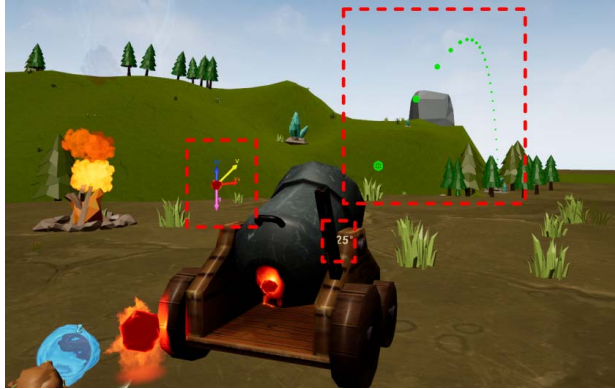


Figure 2: Canon with support elements (a) Shooting angle, (b) vectors, and (c) expected trajectory points

mechanics related to learning, as it allows the player to experience the kinematic phenomena, allowing him/her to adjust the shooting angle, direction, initial speed, among others, as shown in Fig. 2. Other elements support a better visualization of the ball on moving. We can see (a) the shooting angle, (b) vectors that represents the magnitudes of velocities in different axis during fly; and (c) green points that describes the expected trajectory of the ball before the flight.

A game is divided on three levels, each one has a different purpose and variants on the game mechanics. Level 1 is an introductory level, that has a tutorial conducted by a virtual avatar who gives instructions to guide step by step, letting the player be familiarized with the VR environment and controllers. This level has a laboratory teaching function, in case the player wants to try the game for as long as he/she considers necessary. Level 2 is a scenario with static targets that appears in a random location around 360° , at the same ground level, and a 0-500 m distance. Although the objective is simple (hit on these targets), the player has to put into practice some issues like hit same objective with complementary shooting angle, or maximum distance attainable with 45° shooter angle. In Level 3 the cannon is in an above-ground location, and the main task is to hit the moving object that appears in the background and is walking towards the player's location, so the player should consider using the flight time to predict the correct location where the sphere will

collide with the target.

3 CONCLUSIONS AND FUTURE WORKS

This work presented a serious-game based on VR to improve the experimentation and learning of kinematics phenomena. The game was developed using elements of game design theory to propose a complete experience with both games and learning (in other words, to propose game activities in the virtual environment with a strong relationship with the learning goals). Likewise, the VR technology was selected to take advantage of all its benefits such as immersion, engagement and strong relation between the objects behaviour on virtual and real world, that allows simulate physics phenomena as much closer as possible to the real world.

A second release, adapted to standalone devices, has being developed (i.e., a Oculus Quest 2), and also a new game mechanics (changing the gravity, cannon in movements, among others) and topics (statics and dynamics phenomena) could be implemented to expand the scope of the game.

This work is part of a larger research that, besides the focus on assessing immersion and learning gain on students in the field of STEM, also studies the spatial ability as a cross-sectional ability on VR environments.

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